DEPARTMENT OF WORKS & HIGHWAYS
MATERIAL SCIENCE & TECHNOLOGY BRANCH
Materials Testing Laboratory

☑ Port Moresby ☐ Mt. Hagen ☐ Lae ☐ Annex Lab.

Date:

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CALIFORNIA BEARING RATIO - AS1289.6.1.1

Client : Asset No.:

Project: Material Source :

Location: Pit No.:

Material Description: Depth:

Sample Condition (NMC): ....(%)

Soil Type (LL) :.....(%) Others: .....

Curing Time: (Hrs) Initial Time: Date: / / Final Time: Date / /

1	Maximum dry density		t/m^3	
2	Optimum moisture content		%	
3	Mass wet sample		g	
4	Sample moisture content		%	
5	Mass dry sample	(3) x (100 - (4)) / 100	g	
6	Mass of sample at opt.m.c.	(5) x (100 + (2)) / 100	g	
7	Water required/ added	(6) - (3)	ml	
8	..... max. dry density	{(1) x 100… %} / 100	t/m^3	
9	..... % wet density at	{(8) x 100 + (2)} / 100	t/m^3	
10	Mass of wet soil needed	(15) x (9)	g	
11	Mass of wet soil / layer	(10) / ..... Layers	g	
12	Mass os mould and wet soil		g	
13	Mass of mould		g	
14	Mass of wet soil	(12) - (13)	g	
15	Volume of mould		cm^3	
16	Wet density	(14) / (15)	t/m^3	
17	Moisture content		%	
18	Dry density before soak	(16) x100 / (17) +100	t/m^3	
19	Density ratio before soak	(18) x 100 / (1)	%	
20	Dry density after soak	(18) x 100 / (25) + (100)	t/m^3	
21	Density ratio after soak	(20) x 100 / (1)	%	

Expansion				
22	Gauge after soaking		mm	
23	Gauge before soaking		mm	
24	Expansion		mm	
25	Expansion	(24) / 1.17	%	

Absorption				After Soaking	After Test
26	Mass mould & soil after soaking		kg		
27	Mass mould & soil before soaking		kg		
28	Mass of water absorbed	(26) - (27)	kg		

Moisture contentof total sample				After Soaking "B"	After Test "D"
29	Mass wet sample	(14)	g		
30	Mass dry sample	(29) x100 / (17) +100	g		
31	Mass water	(29) - (30)	g		
32	Water absorbed	(28)	g		
33	Total water	(31) + (32)	g		
34	m.c total sample	(33) x100 / (30)	%		

Moisture Content			Before Prep'n	Before Compact'n	'After Compact'n	After test to 30 mm	After test entire depth
Container No.							
Mass container & wet soil	m1	g					
Mass container & dry soil (*)	m2	g					
Mass container	m3	g					
Mass water	m4 = m1 - m2	g					
Mass dry soil	m5 = m2 - m3	g					
Moisture content	w = (m4)/(m5) x 100	%					

(*) See Note 3 in AS.1289.2.1.1

Note: Checked mass (Mc) must be used in all moisture

Tested by: Checked by:

Date: Date:

Authorised by: AS (Material Science & Technology Branch)

FORM No.S-6.1.1 (p1)

Revised Worksheet: 06/04/2022 lhb